High intakes of protein and processed meat associate with increased incidence of type 2 diabetes.

Diets high in protein have shown positive effects on short-term weight reduction and glycaemic control. However, the understanding of how dietary macronutrient composition relates to long-term risk of type 2 diabetes is limited. The aim of the present study was to examine intakes of macronutrients, fibre and protein sources in relation to incident type 2 diabetes. In total, 27 140 individuals, aged 45-74 years, from the population-based Malmö Diet and Cancer cohort, were included. Dietary data were collected with a modified diet history method, including registration of cooked meals. During 12 years of follow-up, 1709 incident type 2 diabetes cases were identified. High protein intake was associated with increased risk of type 2 diabetes (hazard ratio (HR) 1.27 for highest compared with lowest quintile; 95 % CI 1.08, 1.49; P for trend = 0.01). When protein consumption increased by 5 % of energy at the expense of carbohydrates (HR 1.20; 95 % CI 1.09, 1.33) or fat (HR 1.21; 95 % CI 1.09, 1.33), increased diabetes risk was observed. Intakes in the highest quintiles of processed meat (HR 1.16; 95 % CI 1.00, 1.36; P for trend = 0.01) and eggs (HR 1.21; 95 % CI 1.04, 1.41; P for trend = 0.02) were associated with increased risk. Intake of fibre-rich bread and cereals was inversely associated with type 2 diabetes (HR 0.84; 95 % CI 0.73, 0.98; P for trend = 0.004). In conclusion, results from the present large population-based prospective study indicate that high protein intake is associated with increased risk of type 2 diabetes. Replacing protein with carbohydrates may be favourable, especially if fibre-rich breads and cereals are chosen as carbohydrate sources.